

GANESH GUPTA

817-822-1131 | iamgs10rk@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Own technical design end to end for Python (FastAPI) RESTful microservices on Kubernetes with OpenAPI-documented endpoints, sustaining 99%+ uptime at sub-200ms p99 latency on multi-tenant AWS infrastructure.
- Integrated AWS Bedrock and Anthropic Claude foundation models into production Python services for managed, scalable model access alongside GPT and Llama.
- Tuned PostgreSQL query and schema performance and Redis checkpointing for stateful recovery, cutting p99 latency and enabling fault-tolerant workflow resumption.
- Built automated test suites and observability tooling replacing polling with event-driven monitoring, cutting redundant API calls 90% and enabling real-time reliability tracking.
- Drove a deployment-gated evaluation pipeline with CI/CD (Docker, GitHub Actions) enforcing quality and reliability checks, blocking 15% of failed releases before production.

QureAI (AI Startup)

Mar 2026 – Present

ML Infrastructure Engineer

- Built a model observability framework tracking drift, latency, and evaluation regressions in production, enabling governance-aligned rollback workflows as an early engineer at this AI startup.
- Improved RAG retrieval recall 61% to 89% on a 2K-query eval set by rebuilding the chunking pipeline with hybrid dense-sparse retrieval and re-ranking.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production C++ and SQL-backed microservices in distributed systems for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Optimized SQL schema design and query performance, cutting nightly data aggregation from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Standardized failure-handling contracts (retry, timeout, propagation semantics) across 25+ services, reducing production escalations 60% in one quarter.
- Applied OOP principles and design patterns to build testable, maintainable backend components, mentoring engineers through code reviews and design write-ups.

Projects

LangFetch | tinyurl.com/LangFetch

- LangGraph agent compiling natural-language questions into dependency-ordered SQL subqueries with Pinecone vector retrieval and hybrid re-ranking; deployed as an MCP-compatible server for enterprise data queries.

Distributed KV Store / VPC Control Plane Simulator | github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus key-value store in C++ with leader election, log replication, and log compaction; companion VPC control plane simulator modeling subnet routing for fault-tolerant distributed coordination.

Watch-your-LLM | github.com/ganeshhgupta/watch-your-llm

- Production LLM eval platform with hallucination detection, drift tracking, rubric-based scoring, and regression alerting across model versions; FastAPI backend, PostgreSQL.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, Java, C++, TypeScript, SQL

Backend & APIs: FastAPI, Flask, RESTful APIs, OpenAPI, Distributed Systems, Microservices, Testing

AI & Databases: AWS Bedrock, GCP Vertex AI, Gemini, PostgreSQL, DynamoDB, Query/Performance Tuning

Cloud & DevOps: AWS (Lambda, EC2, RDS, SQS), GCP, Kubernetes, Docker, Git, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer